Indian Institute of Technology Kanpur

Department of Space, Planetary & Astronomical Sciences & Engineering (SPASE)

SPA617: Radio Astronomy Laboratory

RF Components: Characterization and Measurement

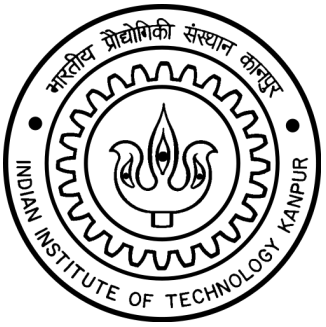
Laboratory Report

Submitted by:

MOHD MEHRAJ
Roll No: 261310610
Ph.D / SPASE

Course Instructor :

Prof. Mugundhan V



Academic Session 2026–2027

1 Introduction and Objectives

Radio-frequency (RF) components are important parts of a radio receiver chain and are used to process the received signals. In this experiment, we study some of these components and observe how they behave when an RF signal is applied to them. The experiment includes a filter, a quarter-wave stub, a mixer, a power splitter/combiner, and an RF amplifier. Through these measurements, the main purpose is to understand the basic working of these components and to measure their performance.

1.1 Aim

The aim of this experiment is to understand the operation and experimentally characterize the filter, mixer, power splitter/combiner, and amplifier used in an RF receiver chain.

1.2 Objectives

The main objectives of the experiment are:

(a) To characterize an RF filter, identify its type, and determine its cutoff frequencies or passband.

(b) To demonstrate the operation of a quarter-wave open-circuit stub as a notch filter.

(c) To observe frequency conversion in an RF mixer and estimate its conversion loss.

(d) To measure the power division and insertion loss of a power splitter/combiner.

(e) To measure amplifier gain, harmonic generation, the 1-dB compression point, and the third-order intercept point (TOI).

(f) To estimate the input reflection coefficient and return loss of the filter.

1.3 Components Required

The following components and instruments were used in the experiment:

- RF signal generator(s)
- Spectrum analyser
- Digital storage oscilloscope
- RF filter
- RF mixer
- Power splitter/combiner
- RF amplifier and DC power supply
- RF Tee connector
- Open-ended coaxial stub
- Short reference coaxial cable
- 50 Ω terminations
- Suitable RF cables, adapters, and attenuators

2 Theory

2.1 RF Receiver Chain

A radio receiver receives signals from the antenna and processes them in several stages before they are used for further analysis. Different RF components are used at these stages because the received signal may need to be amplified, shifted to another frequency, or separated from unwanted signals.

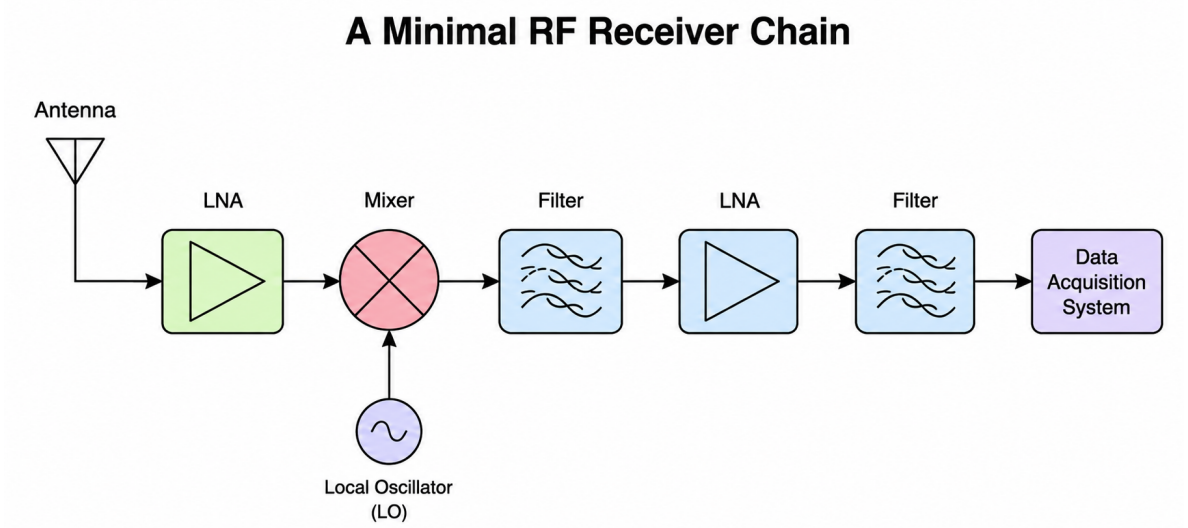


Figure 1: A minimal RF receiver chain showing the main stages from the antenna to the data acquisition system.

An RF receiver chain may therefore contain an LNA, mixer, filter, and power splitter or combiner. The LNA increases the level of a weak received signal, while the mixer changes its frequency. Filters are then used to select the required frequency range and reduce unwanted components. A power splitter or combiner can be used when the signal has to be divided between different paths or when signals from different paths need to be combined.

The arrangement of these components depends on the receiver design. A simple receiver chain is shown in Fig. 1.

2.2 RF Filters

An RF filter is used to control which frequency components of a signal are allowed to pass through a circuit. Its response depends on frequency, so the same input power can produce different output powers at different frequencies. By measuring this response over a range of frequencies, the useful and rejected frequency regions of the filter can be identified.

The commonly used RF filter types are low-pass, high-pass, band-pass, and band-reject (notch) filters.

- A **low-pass filter** allows frequencies below a certain frequency to pass and attenuates frequencies above it.
- A **high-pass filter** attenuates frequencies below its cutoff frequency and allows higher frequencies to pass.
- A **band-pass filter** allows a selected range of frequencies to pass. This range is bounded by a lower cutoff frequency f_1 and an upper cutoff frequency f_2.

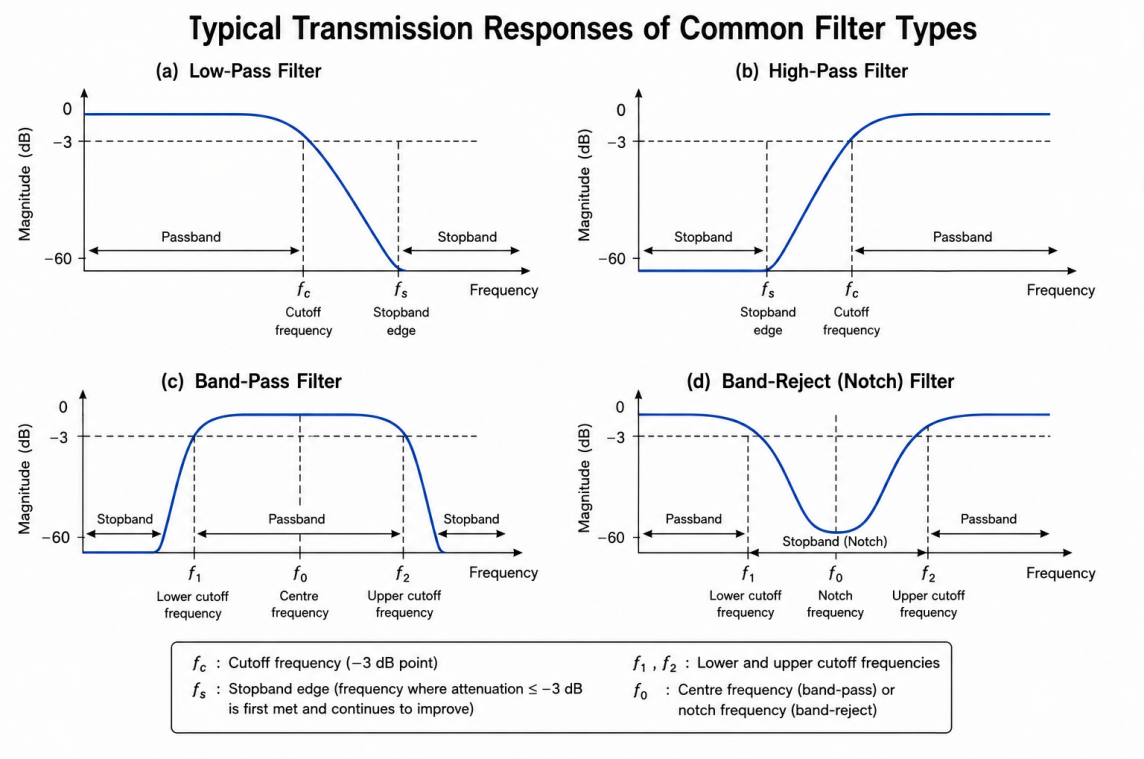


Figure 2: Typical transmission responses of common filter types.

- A **band-reject or notch filter** suppresses a selected frequency range while allowing frequencies outside that range to pass. The frequency of strongest rejection is called the notch frequency f_0.

2.3 Filter Transmission and Insertion Loss

The transmission response of a filter shows how its output changes with frequency. The important frequencies depend on the type of filter. For a low-pass or high-pass filter, the main reference frequency is the cutoff frequency f_c, while f_s represents the stopband edge. For a band-pass filter, f_1 and f_2 are the lower and upper cutoff frequencies. For a band-reject filter, these frequencies mark the boundaries of the rejected region, and f_0 represents the notch frequency.

To measure the transmission of a filter, the power is first measured without the filter. This measured power is taken as P_{in}. The filter is then connected in the same setup and the output power is measured as P_{out}. The difference between these two values gives the insertion loss:

$$IL = P_{\text{in}} - P_{\text{out}} \quad \text{dB}. \tag{1}$$

A smaller insertion loss means that less power is lost while the signal passes through the filter. Therefore, the insertion loss can be used to compare how well the filter transmits signals in its passband.

2.4 Filter Reflection and Return Loss

Not all of the incident RF signal necessarily enters the filter. If the input impedance of the filter is different from the characteristic impedance of the RF system, part of the signal is reflected back towards the source.

Let Z_{in} be the input impedance of the filter and Z_0 be the characteristic impedance of the RF system. The voltage reflection coefficient is given by

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}, \tag{2}$$

where V_{inc} is the incident voltage and V_{ref} is the reflected voltage.

The magnitude $|\Gamma|$ tells us how much of the incident voltage is reflected. For a perfectly matched filter,

$$Z_{\text{in}} = Z_0 \qquad \Rightarrow \qquad |\Gamma| = 0. \tag{3}$$

In this case, there is no reflected signal. On the other hand, if the signal is completely reflected,

$$|\Gamma| = 1. \tag{4}$$

The reflection coefficient is also expressed in terms of return loss:

$$RL = -20 \log_{10} |\Gamma| \quad \text{dB}. \tag{5}$$

A good impedance match gives a small value of $|\Gamma|$ and therefore a large return loss. A larger reflection coefficient gives a smaller return loss.

The limiting cases are summarized below:

Table 1: Relation between reflection coefficient and return loss.

Condition	$\|\Gamma\|$	**Return Loss**
Perfect match	0	∞
Partial mismatch	$0 < \|\Gamma\| < 1$	> 0 dB
Complete reflection	1	0 dB

2.5 Quarter-Wave Open-Circuit Stub Filter

A quarter-wave open-circuit stub can be used to produce a notch in the transmission response. The stub is connected to the RF line through a Tee connector, with its other end left open.

Although the end of the stub is open, the impedance seen at the Tee junction changes with frequency because the signal travels along the transmission line and is reflected from the open end. At a particular frequency, when the electrical length of the stub is approximately one quarter of a wavelength, the open circuit at the far end is transformed into a low impedance at the Tee junction. This strongly reduces the signal passing through the main RF path and produces a notch in the response.

The quarter-wave condition is

$$L = \frac{\lambda}{4}, \tag{6}$$

where L is the physical length of the stub and λ is the guided wavelength in the cable.

The guided wavelength is related to the velocity factor of the cable by

$$\lambda = \frac{VF\, c}{f_0}, \tag{7}$$

where VF is the velocity factor, c is the speed of light in vacuum, and f_0 is the expected notch frequency. Therefore,

$$L = \frac{VF\,c}{4f_0}, \tag{8}$$

or, equivalently,

$$f_0 = \frac{VF\,c}{4L}. \tag{9}$$

The calculated value of f_0 can be compared with the notch frequency measured in the experiment. The two values may not be exactly the same because the connector and the open end of the stub also affect its electrical length.

2.6 RF Mixer

An RF mixer is used to change a signal from one frequency to another. It is a nonlinear three-port device with RF, local oscillator (LO), and intermediate-frequency (IF) ports. When signals with frequencies f_{RF} and f_{LO} are applied, new frequency components are produced at the sum and difference of the two input frequencies:

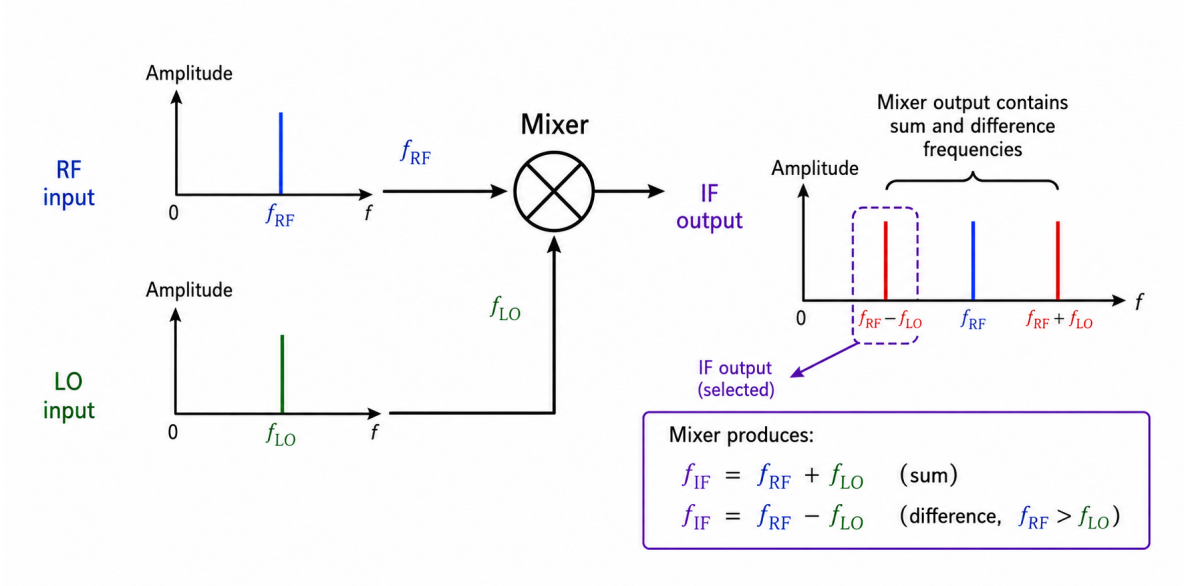


Figure 3: Operation of a mixer.

$$f_{\mathrm{IF}} = f_{\mathrm{RF}} + f_{\mathrm{LO}}, \tag{10}$$

and

$$f_{\mathrm{IF}} = f_{\mathrm{RF}} - f_{\mathrm{LO}}, \qquad f_{\mathrm{RF}} > f_{\mathrm{LO}}. \tag{11}$$

Thus, the output of a mixer can contain more than one frequency component. The required component can be selected according to the receiver design.

The frequency conversion is not ideal, so some power is lost during the conversion. If P_{RF} is the RF input power and P_{IF} is the power of the selected IF component, the conversion loss is

$$\text{Conversion Loss} = P_{\mathrm{RF}} - P_{\mathrm{IF}} \quad \text{dB}. \tag{12}$$

2.7 Power Splitter and Combiner

A power splitter divides the power of one input signal between two or more output ports. For an ideal two-way equal splitter, the input power is divided equally between the two outputs. Each output is therefore

$$3 \text{ dB} \tag{13}$$

below the input power.

For a practical splitter, there is usually some additional loss. If P_{in} is the input power and P_{out1} and P_{out2} are the powers at the two output ports, the losses are

$$Loss_1 = P_{\text{in}} - P_{\text{out1}}, \tag{14}$$

$$Loss_2 = P_{\text{in}} - P_{\text{out2}}. \tag{15}$$

A power combiner performs the reverse function by combining signals applied to its input ports. The resulting output depends on the frequencies, amplitudes, and relative phases of the input signals. For signals with different frequencies, the output contains the corresponding frequency components, while signals at the same frequency can combine differently depending on their relative phase.

3 Experimental Setup and Procedures

The experiment was carried out using an RF signal generator, spectrum analyser, the required RF components, and suitable RF cables and adapters.

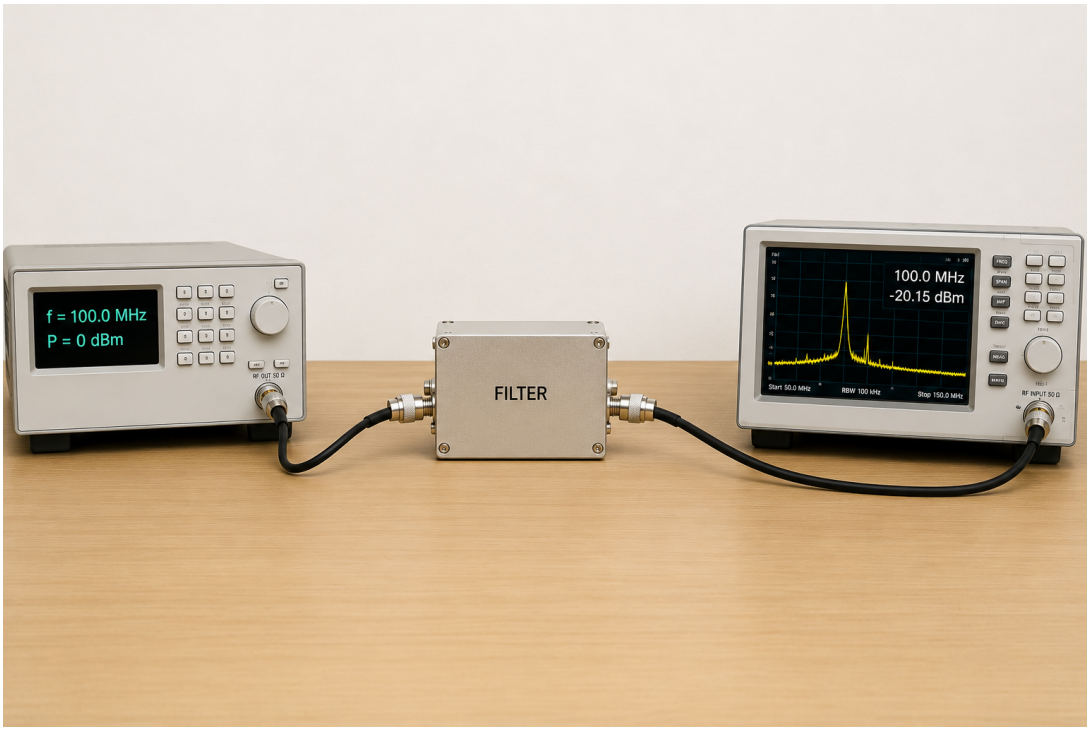


Figure 4: Experimental setup for insertion loss measurement.

For the filter Characterization, the low-pass filter was connected between the RF signal generator and the spectrum analyser. A short reference cable was first used to measure the

power without the filter. The filter was then connected in the same RF path and the output power was measured over the same range of frequencies.
For the quarter-wave stub measurement, the RF signal generator and spectrum analyser were connected through an RF Tee connector. The open-ended coaxial stub was connected to the third port of the Tee connector. Then the output power was measured over the frequency range.

For the mixer measurement, two RF signal generators were connected to the RF and LO ports of the mixer. The IF output was connected to the spectrum analyser. The RF and LO signals were applied at the selected frequencies, and the output spectrum was observed to identify the frequency-converted components.

For the power splitter measurement, the input port of the splitter was connected to the RF signal source and the output ports were measured using the spectrum analyser. The output power at the two ports was recorded to determine the power division and loss.

For the combiner measurement, two RF signals were applied to the input ports of the combiner and the combined output was observed using the spectrum analyser. The output frequency components and their power levels were recorded.

4 Observations, Calculations and Results

The measured values were used to calculate the required parameters for each part of the experiment.

4.1 A1 – RF Filter Characterization

The filter was tested by measuring the output power first without the filter and then with the filter connected. The insertion loss was calculated using

$$IL = P_{\text{without}} - P_{\text{with}}. \tag{16}$$

The measured and estimated reference powers, together with the measured powers with the filter, were used to calculate the insertion loss at each frequency.

Table 2: Measured power response of the low-pass filter.

Frequency (MHz)	P_{without} **(dBm)**	P_{with} **(dBm)**	P_{net} **(dB)**	IL **(dB)**
100	-5.70	-5.49	0.21	-0.21
150	-5.70	-5.73	-0.03	0.03
200	-5.70	-6.06	-0.36	0.36
250	-5.70	-5.94	-0.24	0.24
300	-5.70	-6.43	-0.73	0.73
350	-5.70	-6.62	-0.92	0.92
400	-5.97	-6.81	-0.84	0.84
450	-5.97	-7.40	-1.43	1.43
470	-5.97	-7.54	-1.57	1.57
490	-5.97	-7.56	-1.59	1.59
510	-6.03	-7.78	-1.75	1.75
530	-6.03	-8.46	-2.43	2.43
550	-6.03	-9.40	-3.37	3.37
560	-6.03	-9.27	-3.24	3.24
580	-6.03	-10.96	-4.93	4.93
600	-6.03	-12.24	-6.21	6.21
610	-6.11	-13.40	-7.29	7.29

Frequency (MHz)	P_{without} **(dBm)**	P_{with} **(dBm)**	P_{net} **(dB)**	IL **(dB)**
620	-6.11	-15.42	-9.31	9.31
630	-6.11	-18.52	-12.41	12.41
640	-6.11	-22.62	-16.51	16.51
650	-6.11	-26.75	-20.64	20.64
660	-6.35	-28.49	-22.14	22.14
670	-6.35	-29.22	-22.87	22.87
680	-6.35	-30.45	-24.10	24.10
690	-6.35	-31.33	-24.98	24.98
700	-6.35	-32.96	-26.61	26.61
720	-6.10	-39.78	-33.68	33.68
740	-6.10	-45.86	-39.76	39.76
750	-6.10	-44.40	-38.30	38.30
755	-6.10	-44.09	-38.00	38.00
760	-6.10	-43.86	-37.76	37.76

Cutoff and Stopband Edge Frequencies

The maximum measured net power in the passband is

$$P_{\text{net,max}} = 0.21 \text{ dB}, \tag{17}$$

which occurs at 100 MHz. The cutoff frequency is defined at the point where the transmission decreases by 3 dB from this maximum passband level. Therefore, the cutoff level is

$$P_{\text{net,c}} = P_{\text{net,max}} - 3 = 0.21 - 3 = -2.79 \text{ dB}. \tag{18}$$

From the measured data,

$$P_{\text{net}}(530 \text{ MHz}) = -2.43 \text{ dB},$$

while

$$P_{\text{net}}(550 \text{ MHz}) = -3.37 \text{ dB}.$$

Thus, the cutoff frequency lies between 530 and 550 MHz. Using linear interpolation,

$$f_c = 530 + \frac{-2.79 - (-2.43)}{-3.37 - (-2.43)}(550 - 530). \tag{19}$$

Hence,

$$\boxed{f_c \approx 537.7 \text{ MHz}}. \tag{20}$$

The stopband edge is identified from the measured response as the point where the rapid decrease in transmission has essentially ended and the response begins to approach an approximately constant, strongly attenuated level. From the measured insertion-loss curve, this occurs at approximately 740 MHz. Therefore,

$$\boxed{f_s \approx 740 \text{ MHz}}. \tag{21}$$

The response becomes more nearly constant above approximately 750–760 MHz, supporting the identification of the stopband region.

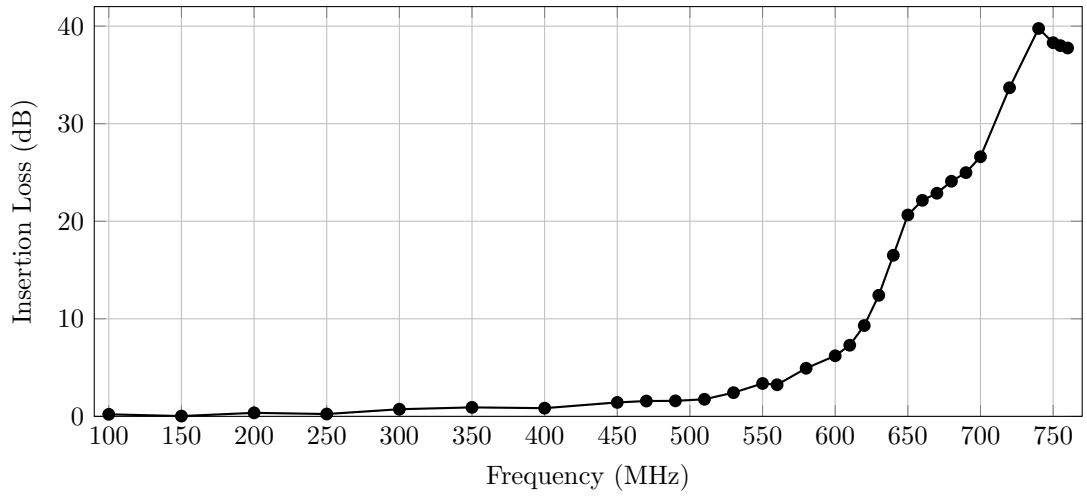


Figure 5: Measured insertion loss of the low-pass filter.

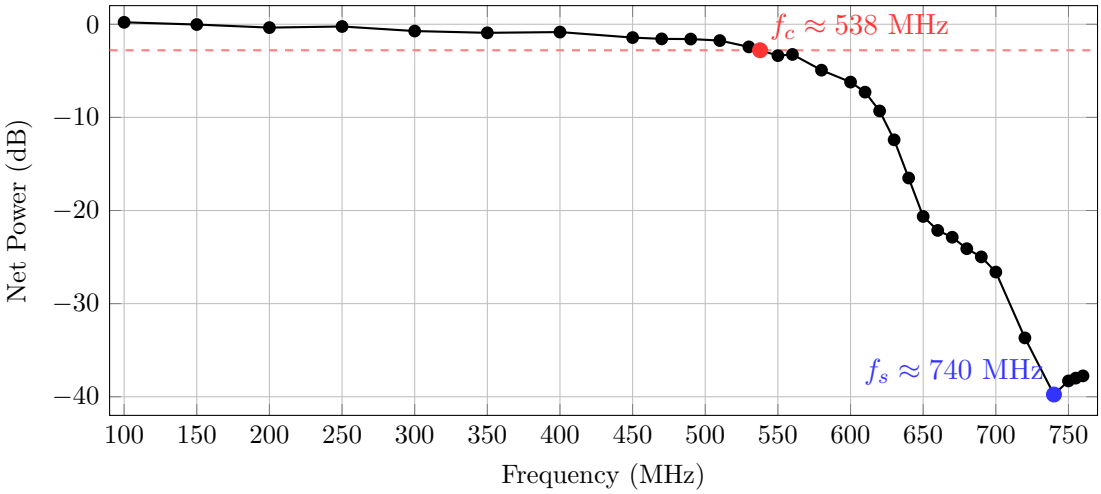


Figure 6: Measured net-power response of the low-pass filter. The cutoff frequency is obtained from the −3 dB level, while the stopband edge is estimated from the point where the rapid attenuation approaches the approximately constant stopband response.

4.2 A3 – Quarter-Wave Open-Circuit Stub

For the open-ended coaxial stub, the expected notch frequency is given by

$$f_0 = \frac{VF\,c}{4L}, \tag{22}$$

where VF is the velocity factor, c is the speed of light, and L is the physical length of the stub.

The recorded values were

$$L = 92 \text{ cm} = 0.92 \text{ m},$$

$$VF = 0.66,$$

and

$$P_{\text{in}} = -3 \text{ dBm}.$$

Therefore,

$$f_0 = \frac{0.66(2.998 \times 10^8)}{4(0.92)}. \tag{23}$$

Hence,

$$\boxed{f_{0,\text{cal}} \approx 53.8 \text{ MHz}}. \tag{24}$$

The attenuation produced by the stub was calculated from

$$A = P_{\text{in}} - P_{\text{out}}. \tag{25}$$

The calculated values are given below.

Table 3: Measured response of the quarter-wave stub.

Frequency (MHz)	P_{out} (dBm)	Attenuation (dB)
10	-5.33	2.33
20	-6.09	3.09
30	-7.29	4.29
40	-9.25	6.25
50	-14.30	11.30
52	-16.50	13.50
54	-20.20	17.20
56	-27.09	24.09
57	-32.47	29.47
58	-30.17	27.17
59	-25.72	22.72
60	-21.60	18.60
62	-17.08	14.08
65	-13.30	10.30
70	-10.12	7.12
80	-7.69	4.69
90	-6.56	3.56
100	-5.80	2.80

The lowest output power occurs at

$$\boxed{f_{0,\text{meas}} \approx 57 \text{ MHz}}, \tag{26}$$

where

$$P_{\text{out}} = -32.47 \text{ dBm}. \tag{27}$$

The maximum measured attenuation is therefore

$$A_{\text{max}} = -3 - (-32.47) = 29.47 \text{ dB}. \tag{28}$$

The difference between the calculated and measured notch frequencies is

$$\Delta f = 57 - 53.8 \approx 3.2 \text{ MHz}. \tag{29}$$

The percentage difference is approximately

$$\frac{57 - 53.8}{53.8} \times 100 \approx 6.0\%. \tag{30}$$

The difference can be associated with practical effects such as connector dimensions and the open-end effect of the coaxial stub.

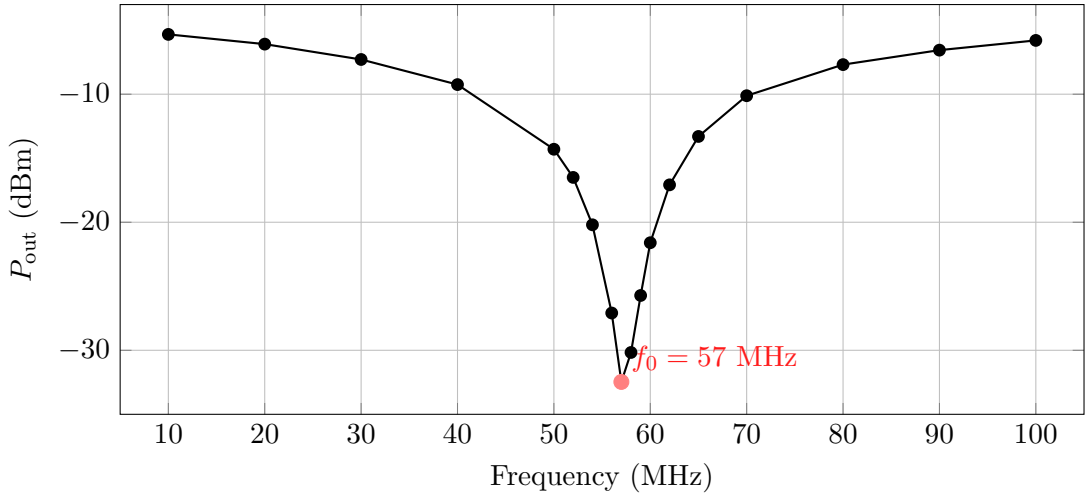


Figure 7: Measured output power of the quarter-wave open-circuit stub as a function of frequency.

4.3 B – Mixer Characterization

The local oscillator frequency was kept at

$$f_{\text{LO}} = 30 \text{ MHz}. \tag{31}$$

The recorded RF input power was

$$P_{\text{RF}} = 0 \text{ dBm}, \tag{32}$$

while the LO power was set to approximately 0 dBm.
For each RF frequency, the expected mixer output frequencies were calculated from

$$f_{\text{IF}} = f_{\text{RF}} - f_{\text{LO}} \tag{33}$$

and

$$f_{\text{IF}} = f_{\text{RF}} + f_{\text{LO}}. \tag{34}$$

The measured powers of the difference- and sum-frequency components were recorded. To obtain a representative IF power for each RF input frequency, the two measured conversion products were averaged:

$$P_{\text{IF,avg}} = \frac{P_{\text{IF,diff}} + P_{\text{IF,sum}}}{2}. \tag{35}$$

The conversion loss was then calculated using

$$CL = P_{\text{RF}} - P_{\text{IF,avg}}. \tag{36}$$

Table 4: Mixer frequency conversion and average conversion loss.

f_{RF} (MHz)	f_{LO} (MHz)	$f_{RF} - f_{LO}$ (MHz)	P_{IF} (dBm)	$f_{RF} + f_{LO}$ (MHz)	P_{IF} (dBm)	CL (dB)
120	30	90	-12.32	150	-12.48	2.40
150	30	120	-12.56	180	-12.50	2.53
180	30	150	-12.40	210	-12.51	2.46

For $f_{RF} = 120$ MHz, the average IF power is

$$P_{IF,avg} = \frac{-12.32 + (-12.48)}{2} = -12.40 \text{ dBm}. \tag{37}$$

Therefore,

$$CL = 0 - (-12.40) = 12.40 \text{ dB}. \tag{38}$$

For $f_{RF} = 150$ MHz,

$$P_{IF,avg} = \frac{-12.56 + (-12.50)}{2} = -12.53 \text{ dBm}, \tag{39}$$

giving

$$CL = 0 - (-12.53) = 12.53 \text{ dB}. \tag{40}$$

For $f_{RF} = 180$ MHz,

$$P_{IF,avg} = \frac{-12.40 + (-12.51)}{2} = -12.455 \text{ dBm}, \tag{41}$$

giving

$$CL = 0 - (-12.455) = 12.455 \text{ dB}. \tag{42}$$

The average conversion loss over the three RF input frequencies is

$$\overline{CL} = \frac{12.40 + 12.53 + 12.455}{3}, \tag{43}$$

giving

$$\boxed{\overline{CL} \approx 12.46 \text{ dB}}. \tag{44}$$

Thus, the mixer produced both the sum and difference frequency components. The average measured conversion loss, based on the average power of the two conversion products, was approximately

$$\boxed{\overline{CL} \approx 12.46 \text{ dB}}. \tag{45}$$

4.4 C1 – Power Splitter

The input power was set to

$$P_{in} = 0 \text{ dBm}. \tag{46}$$

For an ideal two-way splitter, each output should be

$$P_{ideal} = P_{in} - 3 = -3 \text{ dBm}. \tag{47}$$

The measured losses were calculated using

$$Loss_1 = P_{\text{in}} - P_{\text{out1}}, \tag{48}$$

$$Loss_2 = P_{\text{in}} - P_{\text{out2}}. \tag{49}$$

Table 5: Measured power division of the two-way RF splitter.

Frequency (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	$Loss_1$, $Loss_2$ (dB)
150	0	-6.32	-6.19	6.32, 6.19
200	0	-5.85	-5.89	5.85, 5.89
300	0	-6.58	-6.59	6.58, 6.59

The ideal output power for an equal two-way splitter is 3 dB below the input power. For $P_{\text{in}} = 0$ dBm, the ideal output power is therefore

$$P_{\text{out,ideal}} = -3 \text{ dBm}. \tag{50}$$

At 150 MHz, the measured output powers were -6.32 dBm and -6.19 dBm, compared with the ideal value of -3 dBm. At 200 MHz, the measured values were -5.85 dBm and -5.89 dBm, while at 300 MHz they were -6.58 dBm and -6.59 dBm.

The two output powers are close to each other at each frequency, indicating that the splitter provides approximately equal power division between its two output ports. The measured outputs are lower than the ideal -3 dBm value, indicating additional practical loss in the splitter.

4.4.1 Results and Conclusions

For the low-pass filter, the measured transmission response confirmed the low-pass characteristic. The cutoff frequency obtained from the -3 dB point was

$$\boxed{f_c \approx 538 \text{ MHz}}. \tag{51}$$

The stopband edge was estimated from the measured response, following the shape of the reference filter response in the laboratory manual, as

$$\boxed{f_s \approx 740 \text{ MHz}}. \tag{52}$$

For the quarter-wave open-circuit stub, a pronounced attenuation notch was observed near 57 MHz.

For the mixer, both the sum and difference frequency components were observed. The average conversion loss calculated from the measured conversion products was The measured output powers were lower than the ideal -3 dBm value for a 0 dBm input, indicating practical losses in the component.

$$\boxed{\overline{CL} \approx 12.46 \text{ dB}}. \tag{53}$$

For the two-way power splitter, the two output powers were close to each other at all measured frequencies, indicating approximately equal power division. The measured output powers were lower than the ideal -3 dBm output expected for a 0 dBm input.

Sources of Error and Uncertainty

The main sources of error may be because of measurement uncertainty in the spectrum analyser and signal generator, cable and connector losses, impedance mismatch, and small variations in the experimental connections. Frequency resolution and interpolation also introduced uncertainty when determining the cutoff and notch frequencies. These effects can cause small differences between the measured and theoretical results.

Use of Artificial Intelligence

Artificial intelligence tools were used as a learning and writing support during the preparation of this laboratory report. The main uses were:

- To understand the operating principles of RF components, including RF filters, quarter-wave stubs, mixers, power splitters/combiners, and RF transmission measurements.
- To clarify the physical meaning of the equations used in the experiment, particularly insertion loss.
- To organize the experimental procedure and present the laboratory measurements in a clear report structure.
- To assist with LaTeX and Overleaf formatting, including tables, equations, figures, captions, and document layout, help in writing content.
- For calculations and help analyse the measured data, including filter cutoff frequency, stopband behaviour, mixer conversion loss, power division, and attenuation, making that in latex format.
- To help interpret the measured RF responses and compare the experimental results with the expected behaviour described in the laboratory manual.
- To create simple illustrative diagrams of the experimental setups and RF component configurations.